Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	0.801
			Total 1 mark

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses Poisson λ = 10 PI	AO1.1a	M1	$V + W$ is Po(10)
	Obtains correct probability	AO1.1b	A1	P(S > 10) = 1 − P(S ≤ 10) = 0.417
(b)	States that model requires independence of purchases from store to store.	AO3.5b	E1	Purchases of printers at *Verigood* are not independent of those at *Winnerprint*
				Total 3 marks

Q3.

(a) (i) $e^{-1.5} \times 1.5^3 / 3!$

M1

= 0.126

0.125 to 0.126

A1

2

(ii) Using Po(1), P(X < 1) = 1 − P(X ≤ 1)

M1

= 1 − 0.7358 = 0.264

SC Award M1 only if obtain 0.0902 using Po(0.5)

A1

2

(iii) Weekdays Po(7.5) weekend Po(1)

Weekdays = 7.5

M1

Total Po(8.5)

A1

P(Total < 10) = P(Total ≤ 9)

Applied (0.7764, 0.7166, 0.6530 are evidence)

m1

= 0.653

A1

4

(b) Using Total Po from (a)(iii)
P(> 15) = 0.0138, P(> 16) = 0.0066

M1 using their total providing supporting probabilities seen OE use of P(Total ≤ 15 & 16)

M1

So needs 16 tubes

CAO Answer alone scores B2

A1

2

(c) Average rate of failure unlikely to be constant over the course of a day. Very little use of lights over this period.

One mark for any sensible comment

E1

1

[11]

Q4.

(a) (i) $\dfrac{e^{-3.5} \times 3.5^4}{4!}$

M1

= 0.189

AWRT 0.189 Answer only gets B2

A1

2

(ii) Using or stating Po(0.5)

An answer of 0.0144, 0.3935, 0.6065, 0.9098 or 0.9856 implies award of B1 but no further marks

B1

P(≥ 2) = 1 − P(≤ 1)

or

= 1 − 0.9098

M1

= 0.0902

Accept 0.09

A1

3

(iii) Using Po(14)

Sight of 0.1094, 0.1757, 0.9235, 0.9521

B1

P(≤ 19) − P(≤ 10) = 0.9235 − 0.1757

Allow 0.8752 − 0.1185

or 0.9573 − 0.2517 for M1

M1

= 0.7478

AWFW 0.747 to 0.748

A1

3

(b) **GRBs / explosions / events / etc** will be random and / or independent

GRBs / etc short in comparison to observation period (non-overlapping)

For any valid point

E1

1

[9]

Q5.

(a) $\lambda = 6 \times 2.5 =$ <u>**15**</u>

CAO

B1

$P(W \leq 18) =$ <u>**0.819 to 0.82**</u>

AWFW (0.8195)

B1

2

(b) $P(W > w) \leq 0.05$ $P(W \leq w) \geq 0.95$

Implied by a value of 21, 22 or 23

M1

$w =$ ***22***

CAO

A1 2

[4]

Q6.

(a) (i) $P(X \geq 9) = 1 - P(X \leq 8)$

1 − 0.6530 = 0.347 (B1)

= 1 − 0.5231
= 0.4769

awfw 0.476 and 0.477

B2,1 2

(ii) $P(5 < X < 10) = P(X \leq 9) - P(X \leq 5)$
= 0.653 − 0.1496
= 0.5034

awfw 0.503 to 0.504
0.7634 − 0.1496 = 0.613 to 0.614 (B2)
0.6530 − 0.2562 = 0.397 to 0.398 (B2)
0.7634 − 0.2562 = 0.507 to 0.508 (B1)
α − 0.1496 or 0.653 − α (B1) iff $0 < p < 1$

B3,2,1 3

(b) $P(Y < 2) = P(Y \leq 1) = P(Y = 0 \text{ or } Y = 1)$

0.8 to 0.81 (B1)

$= e^{-1.5} + e^{-1.5} \times 1.5$

(both)

M1

[0.2231 + 0.3347]
= 0.5578254
= 0.558

awfw 0.557 to 0.56

A1 2

(c) (i) $\lambda = 8.5 + 1.5 = 10$

Allow P(10) or Po(10)

B1 1

(ii) $P(T > 16) = 1 - P(T \leq 16)$

= 1 − 0.9730

M1

= 0.027

A1

2

(iii) $p = {}^3C_2\ 0.027^2 \times 0.973$

for either term correct

M1

$+\ 0.027^3$

for addition of the two correct terms

M1

= 0.002128 + 0.00001968

= 0.0021 [4 dp]

0.0021 or 0.0022 [iff M1M1 (+ 4dp)]

A1

3

Alternative:

$p = 1 - \mathrm{P}(X \le 1)$

$\mathrm{P}(X = 0) + \mathrm{P}(X = 1)$

$= 0.973^3 + 3 \times 0.973^2 \times 0.027 = 0.921167 + 0.076685$

for either term correct

(M1)

$p = 1 - 0.99785$

for 1 − [sum of two correct terms]

(M1)

= 0.0021

0.0021 or 0.0022 [iff M1M1 (+ 4dp)]

(A1)

(3)

[13]

Q7.

(a) (i) $X \sim \mathrm{P_o}(06)$

$\mathrm{P}(X \le 1) = 0\ 8781$

Awrt 0.878

B1

1

(ii) For matches:

The number of run outs: $Y \sim \mathrm{P_o}(0.15)$

$P(Y \geq 1) = 1 - P(Y = 0)$

$$\left.\begin{aligned} &= 1 - e^{-0.15} \\ &= 1 - 0.8607 \end{aligned}\right\}$$

*must **use** P_o (015).*

M1

= 0 1393

awrt 0.139

A1

$P(X \leq 1 \text{ and } Y \geq 1) = 0\ 8781 \times 0\ 1393$

their (a)(i) × their P(Y ≥ 1)

M1

= 0.122

awrt

A1

4

(b) X and Y are independent.

Number of catches and run outs independent

B1

1

(c) (i) For Season:

$S \sim P_o(9.6)$

Use of λ = 9 6 in correct Poisson expression

M1

$$P(S = 10) = \frac{e^{-9.6} \times 9.6^{10}}{10!}$$

= 0.124

A1

2

(ii) $T \sim P_o(9.6 + 2.4) = P_o(12)$

P_o(12) used or seen

B1

$P(T \geq 15) = 1 - P(T \geq 14)$
$= 1 - 0.7720$
$= 0.228$

(1-0.8444 = 0.155 to 0.156)

B2,1

3

[11]

Q8.

(a) (i) $X \sim \text{Po}(13)$

Both Poisson and $\lambda = 13$

B1

1

(ii) $P(X = 20) = P(X \leq 20) - P(X \leq 19)$
$= 0.975(0) - 0.957(3)$
[allow 0.975 – 0.957]

Must use $\lambda = 13$ otherwise M0A0

M1

$= 0.0177$ (3sf)

AWFW 0.0176 to 0.018

or $P(X = 20) = \dfrac{e^{-13} \times 13^{20}}{20!}$ *M1*

= 0.0177 *A1*

A1

2

(iii) $P(6 \leq X \leq 18) = P(X \leq 18) - P(X \leq 5)$
$= 0.930(2)$

M1

– (0.0107 **or** 0.0259)

M1

$= 0.920$ (3sf)

AWFW 0.919 to 0.92

A1

3

(b) Cars not random
Cars not independent

Allow (number of) cars not random / not independent

Mean and Variance of cars different / not equal

B1 for any one of these 3 statements
*Must indicate a reference to **cars***

B1

Mean / Average / λ / 2.6

Correct comment about value of $\lambda \neq 2.6$

greater / less / smaller / different / variable / not constant / too small / too large

Any combination (one from each group):

Any contextual reason that suggests a change in traffic flow, eg due to: rush hour / congestion / traffic jams / accidents / work traffic / school traffic / peak time

*eg mean greater due to rush hour, **or** λ smaller due to congestion, **or** 2.6 too small due to school traffic*

B1

2

(c) $Y \sim \text{Bin}(20,0.2)$

$P(Y \geq 5) = 1 - P(Y \leq 4)$

or:

$$1 - \begin{pmatrix} 0.01153 + 0.05765 + 0.13691 \\ +0.20536 + 0.21820 \end{pmatrix}$$

$= 1 - 0.6296$

1– 0.6296

(Allow 1– 0.8042 seen for M1)

M1

= 0.37(0) (3sf)

AWFW 0.37 to 0.3704

A1

2

(d) X and Y independent

Any statement which indicates two / both events are independent

B1

$p = 0.0177 \times 0.3704$

[their (c)] × [their (a)(ii)]

M1

= 0.00656 (3sf)

AWFW 0.0065 and 0.0067

A1

3

[13]

Q9.

(a) (i) $\mu = \mathrm{E}(X) = \int_0^\infty kxe^{-kx}\,\mathrm{d}x$

M1

$= \left[-xe^{-kx}\right]_0^\infty + \int_0^\infty e^{-kx}\,\mathrm{d}x$

M1

$= 0 + \left[\dfrac{e^{-kx}}{-k}\right]_0^\infty$

A1

$= \dfrac{1}{k}$

AG

3

(ii) $\int_0^m ke^{-kx}\mathrm{dx} = \dfrac{1}{2}$

Or $\mathrm{F}(x) = 1 - e^{-kx}$

M1

$\left[-e^{-kx}\right]_0^m = \dfrac{1}{2}$

$-e^{-km} + 1 = \dfrac{1}{2}$

$\dfrac{1}{2} = 1 - e^{-km}$

A1

$\mathrm{e}^{-km} = \dfrac{1}{2} \Rightarrow km = \ln 2 \Rightarrow m = \dfrac{\ln 2}{k}$

M1A1

$\ln 2 < 1 \Rightarrow m < \mu$

A1

5

(b) (i) $\mathrm{P}(T = 0) = e^{-\frac{t}{\lambda}}$

B1

1

(ii) **(A)** $\mathrm{P}(T > t) = 1 - \mathrm{F}(t) = e^{-\frac{t}{\lambda}}$

M1

$\mathrm{P}(T < t) = \mathrm{F}(t) = 1 - e^{-\frac{t}{\lambda}} \qquad t \geq 0$

AG

A1

2

(B) $\mathrm{f}(t) = \mathrm{F}'(t) = \dfrac{1}{\lambda}e^{-\frac{t}{\lambda}} \qquad t \geq 0$

M1

$\Rightarrow$ Exponential distribution

A1

2

[13]

Q10.

(a) X = no. with blood disorder

for $X \sim \mathrm{B}(25, 0.7)$

$\mathrm{P}(X > 15) = \mathrm{P}(X \geq 16)$

Alternative:

$X \sim B(25, 0.7)$

$P(X > 15) = 1 - P(X \leq 15)$

$= 1 - 0.18943$

$= 0.81057$

Consider $X' \sim \mathrm{B}(25, 0.3)$ then:

$\mathrm{P}(X \geq 16) = \mathrm{P}(X' \leq 9)$

$= 0.8106$

B3 *$0.81 \leq p \leq 0.811$*

B2 *for $0.902 \leq p \leq 0.9022$*

B1 *for $0.5 \leq p \leq 0.95$*

B3, 2, 1

3

(b) (i) $X \sim \mathrm{P}_0(2.6)$

$\mathrm{P}(X \leq 5) = 0.951$

AWRT

B1

1

(ii) $Y \sim \mathrm{P}_0\,(4.9)$

λ = 4.9 stated or used in poisson expression

B1

$P(Y = 10) = \dfrac{e^{-4.9} \times (4.9)^{10}}{10!}$

s

= 0.0164

AWFW 0.016 to 0.0165

A1

3

(iii) $T \sim P_0(7.5)$

2.6 + (their mean in (ii))

B1ft

$P(T > 16) = 1 - P(T \le 16)$

(for 0.9980)

M1

= 1 – 0.9980

= 0.002

CAO (0.00196)

A1

3

[10]

Q11.

(a) (i) $X \sim Po(7)$

$P(X \le 5) = 0.301$

AWFW 0.300 and 0.301

B1

1

(ii) $P(X = 7) = \dfrac{e^{-7} \times 7^7}{7!}$

M1

= 0.149

A1

$P(X \le 7) - P(X \le 6)$

= 0.5987 – 0.4497 (M1)

= 0.149 (A1)

2

(iii) $0.65 \le p \le 0.66$

P(X ≤ 9) – P(X ≤ 4)

B3

0.72 ≤ p ≤ 0.73 **or** 0.52 ≤ p ≤ 0.53

P(X ≤ 10) – P(X ≤ 4)
P(X ≤ 9) – P(X ≤ 5)

(B2)

0.60

P(X ≤ 10) – P(X ≤ 5)

(B1) **3**

(b) No. telephone calls received per hour
= $Y \sim \mathrm{P}_0\,(0.875)$

B1 **1**

(c) (i) Maximum number = 4

B1 **1**

(ii) P(Y < 4) = P(Y = 0, 1, 2, 3)

$$= e^{-0.875}\left(1+\frac{7}{8}+\frac{49}{128}+\frac{343}{3072}\right)$$

= 0.4169(1 + 0.875 + 0.3828 + 0.1117)
= 0.987740443

Any correct expression (B2)
***or** AWFW 0.987 to 0.988*

B2

P(Y ≥ 4) = 1 – 0.9877

1 – (their P(Y < 4))

M1

= 0.0123

AWFW 0.0122 and 0.0123

A1 **4**

SC
P(Y ≤ 4) = 0.997 to 0.998
***or** any correct expression*

B2

P(Y > 4) = 0.002 to 0.003 M1 A0

EXAM PAPERS PRACTICE

(d) λ probably not constant
The number of calls in any time interval
of 1 hour is likely to vary throughout the day.

'System Down'
$\Rightarrow$ *not independent*

E1

1

[13]

Q12.

(a) (i) $P(X \le 3) = 0.515$

0.5152

B1

1

(ii) $P(Y = 5) = \dfrac{e^{-4.4} \times (4.4)^5}{5!}$

$P(Y \le 5) - P(Y \le 4) = 0.7199 - 0.5512$
correct values seen

M1

$= 0.169$

(0.1687)

A1

2

(b) (i) $T = \text{Po}(8.0)$

B1

X and Y are **independent**
(Poisson random variables)

B1

2

(ii) $P(6 < T < 12) = P(T \le 11) - P(T \le 6)$

M1

$= 0.8881 - 0.3134$

A1

$= 0.575$

(0.5747)

A1

3

(iii) $P(T > 14) = 1 - P(T \leq 14)$

M1

$= 1 - 0.9827$
$= 0.0173$
CAO

A1

$p = (0.0173)^2$
[their $P(T > 14)]^2$

M1

$= 0.0003$ (1 sf)
ft if $0 <$ both p's < 1

A1F

4

(iv) $P(T \leq k) > 0.99 \Rightarrow k \geq 15$
$P(T \leq 15) = 0.9918$
$P(T \leq 14) = 0.9827$

M1

∴ minimum number of devices that Joe should keep in stock = 15

A1

2

[14]

Q13.

(a) (i) $X \sim Po(5.0)$
$\Rightarrow P(X < 4) = P(X \leq 3)$
(0.440 to 0.441) for B1

$= 0.265$
CAO

B2

2

(ii) $Y \sim Po(1.5)$

$$\Rightarrow P(Y = 4) = \frac{e^{-1.5} \times (1.5)^4}{4!}$$

M1

$= 0.0471$
(0.047 to 0.0471)

A1

2

(b) (i) $T = X + Y \sim \text{Po}(6.5)$

B1

$\Rightarrow$ P(T > 5) = 1 – P($T \le 5$)

(1 – 0.2237) ***or*** *(1– 0.5265)*

B1

= 1 – 0.369

= 0.631

B1

3

(ii) $p = {}^8C_7\,(0.631)^7\,(0.369) + (0.631)^8$

ft on their p from (b)(i)

Either part attempted

M1ft

p = 0.11758 + 0.02513

(both parts correct)

A1ft

= 0.143

AWFW 1.142 to 0.143 (CAO)

A1

3

(c) (i) Mean = 8

CAO

B1

Variance = s^2 = 16.9

(sample variance = 15.2)

(AWRT)

B1

2

(ii) Poisson not a good model for data

B1dep

Mean ≠ Variance

B1

2

[14]

Q14.

(a) (i) $X \sim \text{Po}(9\ 0) \Rightarrow$

standard deviation = 3

B1 **1**

(ii) $P(6 < X < 12)$

$= P(X \leq 11) - P(X \leq 6)$

M1

$= 0.8030 - 0.2068$

M1ft

$= 0.5962$

CAO

A1 **3**

(b) (i) $T \sim \text{Po}(11.5)$

CAO

B1 **1**

(ii) $P(T \leq 1) = P(T = 0) + P(T = 1)$

$= e^{-11.5} + 11.5e^{-11.5}$

$= 0.000127$

Use of T = 0 and 1

M1

Substitute correctly into formula

M1

AWFW 0.000126 and 0.00013

A1 **3**

(c) $\bar{x} = 12.0$ and $s^2 = 19.3$ ($s = 4.40$)

$\sigma^2 = 17.4$ ($\sigma = 4.17$)

B1

Mean and variance very different

dep on s^2 (or σ^2)

E1

$\Rightarrow$ Po(12.0) not a suitable model

(dep E1)

E1

3

[11]

Q15.

(a) $P(X < 6) = P(X \le 5)$

M1

$= 0.369$

A1

2

(b) $P(Y \ge 1) = 1 - P(Y = 0)$

M1

$= 1 - e^{-1.5}$

M1

$= 1 - 0.223$
$= 0.777$

A1

3

(c) (i) $X + Y \sim \text{Po}(8)$

$$\frac{e^{-8} \times 8^6}{6!}$$

M1

$P(X + Y = 6) = 0.122$

0.3134 – 0.1912

M1A1

3

(ii) $P(Y \ge 1 \mid X < 6) = P(Y \ge 1) = 0.777$

B1

1

[9]

Q16.

(a) $P(X = 8) = P(X \le 8) - P(X \le 7)$
$= 0.8472 - 0.7440$

$P(X = 8) = \dfrac{e^{-6}(6^8)}{8!}$

M1

= 0.103

A1

2

(b) (i) $\lambda = 9$

B1

1

(ii) $P(Y > 9) = 1 - P(Y \le 9)$
= 1 – 0.5874

M1

= 0.4126
AWFW 0.412 to 0.413

A1ft

2

(c) (i) $T \sim Po(15)$

B1ft

1

(ii) $P(T \le 20) = 0.917$

B1ft

1

(iii) $P(T$ at least 21) = 0.083

B1ft

$p = 15 \times (0.083)^4(0.917)^2$
B(6, (iii)) used

M1

= 0.000599
CAO; AWFW 0.000598 to 0.0006

A1

3

[10]

Q17.

(a) (i) $X \sim \text{Po}(0.70)$

$P(X < 3) = P(X \le 2)$

$= 0.966$

0.9659

B1

1

(ii) $Y \sim \text{Po}(1.30)$

$$P(Y = 2) = \frac{e^{-1.30}(1.30)^2}{2!} = 0.230$$

M1A1

2

(b) $T \sim \text{Po}\ (2.0)$

M1

$P(T \ge 4) = 1 - P\ (T \le 3)$

$= 1 - 0.8571$

M1

$= 0.143$

0.1429

A1

3

[6]

Q18.

(a) (i) $$P(A = 4) = \frac{e^{-3.5} \times (3.5)^4}{4!} = 0.189$$

M1A1

2

(ii) $P(B \le 6) = 0.762$

B1

1

(iii) $T = A + B \sim \text{Po}(8.5)$

$P(T \text{ fewer than } 10) = P(T < 10)$

Use of Po(8.5)

M1

$= P(T \le 9)$

$T \le 9$ attempted

M1

= 0.653

CAO

A1

3

(b) X ~ B (5, 0.653)

X ~ B(5, their p)

B1

$$P(X \geq 4) = \binom{5}{4}(0.653)^4(0.347) + (0.653)^5$$

M1

= 0.31547 + 0.11873
= 0.434

On their p from (a)(iii)

A1ft

3

(c) (i) $\bar{x} = 9.2$

B1

$s^2 = 9.29$

$\sigma^2 = 8.36$

B1

2

(ii) Mean and variance have similar values which suggests that Poisson distribution may be appropriate

B1ftB1ft

2

[13]

Q19.

(a) (i) $$P(X = 3) = \frac{e^{-3.5} \times (3.5)^3}{3!} = 0.216$$

M1A1

2

(ii) P($Y \geq 5$) = 1 – P($Y \leq 4$)

used

M1

= 1 – 0.2851
= 0.715

A1

2

(b) (i) $T \sim \text{Po}(9.5)$

B1

1

(ii) $P(7 \le T \le 10) = P(T \le 10) - P(T \le 6)$

M1

= 0.6453 – 0.1649

A1

= 0.408
Accept 0.48

A1

3

(iii) $p = (0.4804)^3 = 0.111$

M1A1ft

2

[10]

Q20.

(a) (i) $P(X \le 1) = 0.9098$

M1

AWRT 0.91

A1

2

(ii) $P(Y = 2) = \dfrac{e^{-2.5} \times (2.5)^2}{2!}$

M1

= 0.257
AWFW 0.256 and 0.257

A1

2

(b) (i) $T = X + Y \sim P_o\,(3.0)$
P(Total of 2 breakdowns) = p([$_{x,y}$] = 2)

= P([2,0]or[0,2]or[1,1])

B1

$$P(T = 2) = \frac{e^{-3.0} \times (3.0)^2}{2!}$$

= [0.0758 × 0.0821] + [0.6065 × 0.25652]
+ [0.20521 × 0.3033] = 0.006223 + 0.155585

M1

= 0.224
+ 0.062233
= 0.224

A1

3

OR

Using tables: = 0.4232 – 0.1991
= 0.224

(ii) $p = (0.224)^4 = 0.0025$

M1 A1

2

(c) (i) $4X \sim P_o(2.0)$ $\Rightarrow$ $\mu = 12.0$
$4Y \sim P_o(l0.0)$

B1

1

(ii) P(T ≥ 18) = l – P(T ≤ 17)

M1

= 1 – 0.9370
= 0.0630

A1

2

[12]

Q21.

(a) (i) $$P(X = 2) = \frac{e^{-1.5} \times (1.5)^2}{2!} = 0.251$$

M1A1

2

(ii) $p = (0.251)^3 = 0.0158$
on their p from (i)

M1A1ft

2

(b) (i) $Y \sim P_o(9.0)$

B1

1

(ii) $P(Y \geq 12) = 1 - P(Y \leq 11)$

$= 1 - 0.8030$

M1

$= 0.197$

A1

2

(c) attacks patients:
randomly (*p* constant)

mean of 1.5 $\Rightarrow$ *p small (B1)*
(unless very few patients)

B1

independently

B1

[9]

Q22.

(a) For a 1-year period
The number of A grades ~ Po (3)

For a 5-year period
The number of A grades ~ Po (15)

B1

P (Total A-grades > 18)
= 1– (Total ≤ 18)

M1

= 1– 0.8195
= 0.1805
= 0.181

AWFW 0.180 and 0.181

A1

3

(b) (i) $X + Y \sim P_o\ (10)$

B1

$P(X+Y \le 14) = 0.917$

(0.916 to 0.917)

M1A1 **3**

(ii) *X* and *Y* are independent variables

E1 **1**

[7]

Q23.

(a) (i) $P(Y = 2) = \dfrac{e^{-1.9} \times (1.9)^2}{2!}$

M1

$= 0.270$

AWRT

A1 **2**

(ii) $(0.270)^5 = 0.00143$

On their (a)(i)
AWRT

M1A1ft **2**

(b) (i) $X \sim P_o(9.5)$

*Poisson **and** 9.5*

B1 **1**

(ii) $P(X \ge 10) = 1 - P(X \le 9)$

M1

$= 1 - 0.5218$
$= 0.4782$

A1 **2**

(iii) $\therefore p = 10 \times (0.4782)^3 (0.5218)^2$

On their b (ii)

M1

$= 0.298$

AWRT 0.3

A1

2

[9]

Q24.

(a) P (X = 2) = $\frac{e^{-2.6}(2.6)^2}{2!}$

0.5184 – 0.2674 = 0.251

M1

= 0.251

A1

2

(b) (i) $Y \sim P_o$ (13)

*Poisson **and** 13*

B1

1

(ii) P ($Y \geq 15$) = 1 – P ($Y < 14$)

M1

= 1 – 0.6751
= 0.3249
= 0.325

On their λ from b (i)

A1ft

$\therefore$ $p = (0.3249)^4$

M1

p = 0.0111 to 0.0112

On their p ($Y \geq 15$)

A1ft

4

[7]

Q25.

(a) (i) $X \sim P_o$ (4.0)

P($X > 8$) = 1 – P($X \leq 8$)

M1

= 1 – 0.9786

= 0.214

(0.021) accept

A1

2

(ii) $X \sim P_0\,(3.5)$

$P(Y < 2) = e^{-3.5}(1 + 3.5)$

M1

= 0.136

(0.13589)

A1

2

(b) (i) $\lambda = E(T) = 7.5$

B1

1

(ii) $P(T \geq 11) = 1 - P(T \leq 10)$

M1

= 1 – 0.8622

= 0.1378

(on their λ)

A1f.t.

2

[7]

Q26.

X = no. of strikes per game

$\therefore\ X \sim P_0(\lambda)$

$X_1 \sim P_0(0.2)$

$X_2 \sim P_0(1.1)$

$X_3 \sim P_0(1.1)$

$\therefore\ T = X_1 + X_2 + X_3 \sim P_0(2.4)$

Adding 3 values must be $P_0(2.4)$

M1A1

$P(3 \leq T \leq 5) = P(T = 3 \text{ or } 4 \text{ or } 5)$

Seen/attempt for $P_0(\lambda)$ only

M1

$= P(X \leq 5) - P(X \leq 2)$

(Poisson tables only)

M1

= 0.9643 – 0.5697
= 0.3946

AWRT 0.395

A1

5

[5]

Q27.

(a) (i) λ = 3.6 for a 5–minute period
let Y = no. of vehicles arriving in a 5 minute period
then $Y \sim P_o(3.6)$

$P(Y \le 2) = 0.3027$
$P(Y \ge 3) = 1 - P(Y \le 2)$

0.303 (AWRT)

Use of tables or $P(Y = 0) = e^{-3.6} = 0.2732$

B1/M1

= 1 – 0.3027

$P(Y = 1) = 3.6\, e^{-3.6} = 0.09837$

= 0.6973
= 0.697

$P(Y{=}2) = \dfrac{(3.6)^2 e^{-3.6}}{2!} = 0.17706$

A1

3

(ii) p = 0.6973 (or 0.697)

<u>use</u> of their $P(Y \ge 3)$

M1

P(at least 3 arrive in 3 succ.5 m)
= $(0.6973)^3$

p^3, iff first M1 gained

m1

= 0.339046
= 0.339

CAO

A1

3

(b) Let X = # vehicles arriving in a 10 - minute period
then $X \sim P_o(7.2)$

FOR 7.2

B1

$P(X = 0) = e^{-7.2}$
$= 0.0007$

$$\begin{cases} e^{-3.6} \times e^{-3.6} \\ \textit{for Poisson, correcly} \\ AG(0.0007465) \end{cases}$$

M1A1

3

[9]

Q28.

(a) (i) P(2 or fewer) = 0.9526

0.953 (0.952 ~ 0.953)

B1

(ii) P (>1,<3) = P(2)

P(2) required

M1

= P (2 or fewer) – P (1 or fewer)

P(2 or fewer) – P(1 or fewer) or correct use of formula

m1

= 0.9526 – 0.8088 = 0.144

0.144 (0.143 ~ 0.1445)

A1

4

(b) (i) mean 5 × 0.8 = 4

4 cao

B1

standard deviation √4 = 2

M1

√ their mean – allow variance = their mean – variance must be stated
2 cao

A1

3

(ii) P (fewer than 3) = 0.2381

(0.2375 – 0.2385)

B1

1

[8]

Q29.

(a) (i) P(3) = 0.9212 – 0.7834 0.138

P (3) = P (3 or fewer) – P (2 or fewer) or use of correct formula

M1

0.138 (0.1375 to 0.1385)

A1

2

(ii) Poisson mean 2.4

Poisson, mean 2.4

B1

P (0) = 0.0907

0.0907 (0.0907 to 0.09075)

B1

2

(iii) Poisson mean 12

Poisson mean 12

B1

P (20 or more) 1 − P (19 or fewer)

P (20 or more) 1 − P (19 or fewer)

M1

= 1 – 0.9787

=0.0213

A1

3

0.0213 (0.021 to 0.0214)

***sc** allow B2 for 0.0116 (0.011 to 0.0117)*

(b) (i) Poisson mean 1.8

Standard deviation $\sqrt{1.8} = 1.34$

ft their mean

M1

1.34 (1.34 to 1.345)

A1

2

(ii) Cannot distinguish between 2 – 1 and 3 – 0

Reason

E1

1

(c) Mean not constant

Reason - generous

E1

1

[11]

Q30.

(a) (i) P(6 or fewer) = 0.3782

Correct use of Poisson tables or formula

0.378 (0.378 – 0.379)

M1 A1

(ii) P(8) = 0.6620 – 0.5246 = 0137

P(8) = P(8 or fewer) – P(7 or fewer)

or *use of correct formula*

0.137 (0.137, 0.138)

M1 A1

4

(b) $\sqrt{7.5}$ = 2.74

method ***sc*** *B1 var = 7.5*

2.74(2.73, 2.74)

M1 A1

2

[6]

Q31.

(a) Poisson

cao

B1

1

(b) (i) P(2 or fewer) = 0.570

0.570 (0.569 to 0.57)

B1

(ii) P(4) = 0.9041 – 0.7787

P(4) = P(4 or fewer) – P (3 or fewer) or correct use of Poisson formula

M1

= 0.125

0.125 (0.125 to 0.126)

A1

3

(c) (i) Poisson, mean 12

Poisson, mean 5 × 2.4 (may be implied)

M1

Poisson mean 12 (may be implied)

A1

P(fewer than 6) = 0.0203

0.0203 (0.02 to 0.021)

B1

(ii) P(> 17) = 1 – 0.9370 = 0.0630

P(>17) = 1 – P(17 or fewer)

M1

0.063 (0.0625 to 0.0635)

A1

5

[9]

Q32.

(a) (i) 0.2600

0.2600 (0.2595 to 0.2605)

B1

(ii) P(14) = 0.5704 – 0.4644 = 0.106

P(14)= P(14 or fewer) – P(13 or fewer)

or correct use of formula

M1

106 (0.1055 to 0.1065)

A1

3

(b) 1 – 0.8272 = 0.173

1 – P(17 or fewer)

M1

0.173 (0.172 to 0.173)

A1

2

[5]

Q33.

(a) 0.463

0.463 (0.462 – 0.463)

B1

1

(b) 0.8913 - 0.7306 = 0.161

M1

P (3) = P(3 or fewer) or
attempted use of correct formula.

M1

Correct use of Poisson tables or
completely correct method
of calculation

A1

3

0.161 (0.16 – 0161)

[4]

Q34.

(a) (i) 0.9526

0.9526 (0.952 – 0.953)

B1

1

(ii) 1 – 0.4493 = 0.5507

P(at least once) = 1 – P(0) or use of correct formula
0.5507 (0.55 – 0.551)

M1 A1

2

(iii) 0.8088 – 0.4493 = 0.3595

P(1) = P(1 or fewer) – P(0) or use of correct formula
0.3595 (0.359 – 0.36)

M1 A1

2

(b) (i) Poisson mean 4

Po(4) may be implied later but not if answered incorrectly

B1

1

(ii) 1 – 0.6288 = 0.3712

1 – P(4 or fewer)
0.3712(0.371 – 0.372)

M1 A1

2

(iii) $0.5507^5 = 0.0506$

Use of their (a)(ii)

[their P(at least once)]^5

0.0506(0.0506 – 0.051)

M1 M1 A1

3

[11]

Q35.

(a) 0.977

Correct use of tables/formula

0.977(0 976 – 0 977)

M1/A1

2

(b) Poisson mean 3

Mean 3

B1

P(>4) = 1 – 8153

Method , allow 1 – 3 or fewer

M1

= 0.185

0.185 (0.185 – 0.186)

A1

3

(c) Individuals do not enter independently

E2(1) independent

E2(1)

2

[7]

Q36.

(a) (i) 0.231

0.23 – 0.232

B1

1

(ii) 1 – 0.4695

P(3 or more) = 1 – P(2 or fewer) or

correct method of calculation

M1

= 0.5305

0.53 – 0.531

A1

2

(b) Poisson mean 5 × 2.8

completely correct method

M1 M1

P(<10) = 0.109

0.109 – 0.11

A1

3

(c) (i) x = 1.3

cao

B1

S^2 = 3.34

3.34 – 3.35

allow 3.01 (3.005 – 3.015)

B1

2

(ii) Mean and variance different

B1

1

(iii) Mean unlikely to be constant over 24 hour period/not independent

B1

1

[10]

Q37.

(a) (2 or fewer) = 0.8335

Correct use of tables or formula

0.833-0.834

M1 A1

2

(b) (4) = 0.9857 – 0.9463 = 0.0394

(4) = (4 or fewer) – (3 or fewer) ***or***

correct use of formula

0.039 – 0.040

M1 A1

2

(c) P(2 or more) = 1 – P(1 or fewer)

= 1 – 0.5918

Completely correct method

M1

= 0.408

0.408 (0.408 – 0.409)

A1

2

[6]

Q38.

(a) (i) P(2 or fewer) = 0.8335

0.8335(0.833 – 0.834)

B1

(ii) Poisson mean 2.8

Poisson mean 2 × 1.4

M1

P(4 or more) = 1 – 0.6919

P(4 or more) = 1 – P(3 or fewer)
Method

M1 M1

= 0.308

A1

5

0.308(0.3075 – 0.3085)

(b) Poisson mean 2.4

M1

P(<4) = 0.7787

A1

2

0.779(0.778 – 0.779)

(c) Poisson mean 3.8

Poisson mean 1.4 + 2.4

M1

$P(> 6) = 1 – 0.9091$
$P(> 6) = 1 – P(6$ or fewer$)$

m1

= 0.0909
0.0909(0.908 – 0.091)

A1
3

(d) Not independent/ mean may vary according to day of the week

E1
1

[11]

Examiner reports

Q1.

The majority of students scored the mark for this question. The most common incorrect answers were P(X ≤ 1) = 0.199 or P(X ≥ 1) = 0.950.

Q3.

This question was generally very well-answered, although a small proportion used incorrect values of λ and scored poorly. In the final part, some students merely used 'standard' Poisson phrases such as 'not independent' or 'mean not equal to variance', but most 'considered the context and produced a sensible explanation.

Q4.

Part (a)(i) was generally well answered and, in parts (ii) and (iii), the value of λ was usually correctly calculated, but the boundary values to be used caused more difficulty. In part (b), answers such as "The days are independent" or "The satellites are independent" showed why it could not be assumed that "They are independent" referred to the GRBs. A not uncommon answer was "Because mean and variance are equal" which showed that students were quoting learnt answers, without any regard for the information given, or not given, in the question.

Q5.

A small minority of students used the normal approximation to the Poisson in part (a) and so scored minimal marks. The majority, who used Poisson tables or their calculators' equivalent, often scored at least 3 marks with the usual mark lost for quoting 21 or 23 instead of 22 in (b).

Q6.

Many fully successful solutions were seen to part (a). In part (b), some students were unable to cope with the fact that the value of λ = 1.5 was not given in Table 2 of the supplied booklet. Thus some tried to use this table by finding the average of the values of P (X ≤ 1) for λ = 1.4 and λ = 1.6. This incorrect method gained no credit. Answers to part (c)(i) were almost always correct as were those to part (c)(ii), except for the students who thought that they needed to evaluate 1 − P (X ≤ 15). The great majority of students realised that the binomial distribution was required in part (c)(iii). Unfortunately, one of the two required terms was often missing.

Q7.

Part (a)(i) was answered very well. However, in part (a)(ii), some candidates attempted to use Po(1.5) or, because Po(0.15) was not in the booklet provided, attempted to average the values found under Po(0.10) and Po(0.20). Neither of these approaches gained any credit. In part (b), it was expected that candidates would state either that 'X and Y are independent' or that 'the number of catches and run-outs are independent'. Although these were often stated, there were candidates who thought that stating 'they are independent' or that 'cricket matches are independent' would suffice. These statements were not sufficient to gain any credit.

Parts (c)(i) and (c)(ii) were usually very well answered. However, some candidates seemed to have some difficulty interpreting statements such as 'at least 15'.

Q8.

In part (a)(i), candidates were expected not only to give the value of λ to be 13 but also to state that X followed a Poisson distribution; this was not always seen. Part (a)(ii) was usually done well either by using tables or by applying the correct formula. Unfortunately some candidates used λ = 2.6. In part (a)(iii), tables were often used correctly and accurately to obtain the correct answer of 0.920. Part (b) was not done at all well, with only the most able candidates gaining any credit. Although many thought that the value of λ would be different from 2.6, they could not give a convincing reason why this should be the case.

Comments such as 'It is not random' or 'Mean not equal to the variance' by themselves were not specific enough and consequently gained no credit. Part (c) was usually very well answered by the vast majority of candidates who used tables. Those who chose to use the formula approach usually either missed one of the terms out or made some error in their calculations. In part (d), the necessary assumption that was expected was 'X and Y are independent'. Comments such as 'They are independent' or 'The speeds are independent' gained no credit. The calculation of the required probability was usually attempted correctly.

Q9.

The method marks in parts (a)(i) and (ii) were generally earned. Almost inevitably there were errors with signs, integration and use of logarithms, costing accuracy marks. Most candidates stated the correct probability in part (b)(i), but failed to recognise that it was needed in part (b)(ii)(A). The hope that 2 marks could be gained by writing down what was on the question paper was forlorn. Candidates needed to state that P(T > t) was the probability that 0 strikes occur in the interval (0, t) and hence that P(T > t) = F(t) = 1 – $e^{t/\lambda}$ for $t \geq 0$. Most candidates knew that they needed to differentiate F with respect to t in order to obtain f(t), but some then forgot to state the distribution of T, which was asked for in the question.

Q10.

Only the most able candidates managed to find the solution to part (a) by stating correctly that P(X > 15) = 1 – P($X \leq$ 15) = 1 – 0.18943 = 0.811. For those using tables for B(25, 0.3), the most popular misconceptions were to obtain P($X \leq$ 15) = 0.9995, and then **either** thinking that this gave the correct answer **or** that, since the value of 0 3 p – = had been obtained by subtracting 0.7 from 1, the correct answer was found by subtracting 0.9995 from 1. Very few candidates were able to use the binomial tables correctly and so P(X > 15) = P($X' \leq$ 9) = 0.8106 was very rarely seen.

Answers to part(b)(i) were usually correct as were those to part (b)(ii), except by those candidates who, on not finding λ = 4.9 in the tables, incorrectly thought that the use of λ = 5 was close enough. In part (b)(iii), candidates were asked to write down the **distribution** of the random variable T. Although the majority stated that λ = 7.5 they did not state that the distribution was Poisson and consequently lost a mark. There were many fully correct solutions to P(T > 16) with most candidates realising that P(T > 16) = 1 – P($X \leq$ 16) = 0.002.

Q11.

The majority of candidates gained full marks for parts (a)(i) and (a)(ii). However, in part (a)(iii), although there were many correct answers seen, there were far too many candidates who could not interpret correctly the expression ‘at least 5 but fewer than 10’.

In part (b), candidates were asked to write down the **distribution** of the number of telephone calls received each hour. Although the majority stated that $\lambda = 0.875$, they did not state that the distribution was Poisson and consequently lost a mark. Part (c)(ii) was answered correctly by, in the main, only the most able candidates. Some who correctly realised that $P(Y \geq 4) = 1 - P(Y \leq 3)$ was required then omitted $P(Y = 0)$ when evaluating $P(Y \leq 3)$ or used $\lambda = 7$ in their calculations. The reasons given in part (d) were not always expressed well enough to gain credit.

Q12.

This was another question which provided many candidates with a good source of marks. Most were able to gain the correct answers to part (a). In part (b)(i), the great majority of candidates realised that **both** Poisson and $\lambda = 8$ were required. When stating their assumption, some candidates simply stated ‘they are independent’ without any actual indication of *what* was independent; this did not gain any credit. In part (b)(ii), whilst most candidates realised that $P(6 < X < 12) = P(X = 7, 8, 9, 10, 11)$ and consequently went on to evaluate $P(X \leq 11) - P(X \leq 6)$ using the tables provided to gain the correct answer, some seemed less certain as to what values were included in $6 < X < 12$. In part (b)(iii), the great majority of candidates correctly translated ‘exceed 14’ and so attempted to find $P(T > 14)$ by evaluating correctly $1 - P(T \leq 14)$. This was invariably followed correctly by $p = 2 \times P(T > 14)$ (1sf).

Unfortunately some candidates thought that $P(T > 14) = 1 - P(T \leq 13)$ and that $p = 2 \times P(T > 14)$. Candidates should realise that where a required degree of accuracy is stated in the question, then they must adhere to this request. Many failed to do this, giving their answer as 0.000299, and consequently lost some credit. In part (b)(iv), there were many good solutions seen with the correct answer of 15 often quoted.

Q13.

There were many fully-correct solutions seen to this question. The vast majority of candidates used the tables provided to enable them to find the correct answer to part (a)(i) by the most efficient method. A minority of candidates seemed only able to use the formula and consequently worked out the individual probabilities before obtaining their answer. In part (a)(ii), the formula was often used correctly to obtain the correct answer of 0.0471. However, there were some candidates who seemed to think that they had to use tables whatever the value of λ and so attempted to average the values found in the tables under $\lambda = 1.4$ and $\lambda = 1.6$ incorrectly thinking that this would give them the required value for $\lambda = 1.5$.

In part (b)(i), most candidates interpreted ‘more than 5’ correctly using $P(X + Y > 5) = 1 - P(X + Y \leq 5)$, whilst others incorrectly used $1 - P(X + Y \leq 4)$ or $1 - P(X + Y \leq 6)$. Only the most able candidates then went on to use their answer to correctly determine the numerical solution to part (b)(ii). There were many erroneous attempts at using various expressions with $0.631^7 + 0.631^8$ or $0.631^7 \times 0.369$ or even $6.5 \times 0.631^7 \times 0.369$ seen, rather than the required $p = {}^8C_7(0.631)^7(0.369) + (0.631)^8$. In part (c)(i), the vast majority of candidates correctly stated that $\bar{x} = 8$ and that $s^2 = 16.9$, but with some stating the sample variance as 15.2. These answers usually enabled candidates to indicate correctly in part (c)(ii) that a Poisson distribution was probably not a good model since the mean and the variance had very different values.

Q14.

In part (a)(i), there were surprisingly many candidates who did not know the relationship between the value of λ and the standard deviation of a Poisson distribution. In part (a)(ii), although the vast majority realised that the most efficient method of answering this part of the question required the use of the table on page 23 of the formulae booklet, very many failed to look up the appropriate values to enable them to show that $P(6 < X < 12) = P(X \le 11) - P(X \le 6) = 0.596$. Those candidates who calculated the separate probabilities for X = 7, 8, 9, 10 and 11 often reached the required answer.

In part (b)(i), it should be noted that $\lambda = 11.5$ and an indication that the distribution of T is Poisson were both required to gain the mark. In part (b)(ii), most candidates realised that "fewer than 2" implied that $P(T \le 1) = P(T = 0) + P(T = 1)$. The correct answer of 0.000127 was then usually found by the use of $\lambda = 11.5$ and the formula $P(X = x) = e^{-\lambda}\dfrac{\lambda^x}{x!}$ (found on page 11 of the formulae booklet).

Although part (c) asked candidates to calculate appropriate numerical measures (plural), this was ignored by some candidates. Such candidates usually calculated $\bar{x}$ and, finding this to coincide with the given value of λ, stated inappropriately that Po(12) was a suitable model for the given data. Of those candidates who did calculate more than one numerical measure, most correctly calculated the mean and variance of the given data ($\bar{x} = 12$; $s^2 = 19.3$ or $\sigma^2 = 17.4$), and then indicated that, because the mean and variance were (very) different values, Po(12) was not a suitable model for the number of places booked each week. Some candidates incorrectly thought that the mean and the standard deviation had to be the same.

Q15.

The majority of candidates managed to complete correctly parts (a), (b) and (c)(i) but some could only give the correct answer to part (a), apparently finding the remainder of the question beyond them. In part (c)(ii), all candidates failed to realise that $P(Y \ge 1 | X < 6) = P(Y \ge 1)$, and consequently no correct answers were seen.

Q16.

Part (a) was answered well by most candidates. The use of tables to evaluate $P(X = 8) = P(X \le 8) - P(X \le 7)$ or the use of the formula $P(X = 8) = \dfrac{e^{-6} \times 6^8}{8!}$ were equally successful methods used here. In part (b)(i), although the correct answer of $\lambda = 9$ was often seen, it was common to see the wrong answer of $\lambda = 3$. In part (b)(ii), the most prevalent error was made by candidates who wrongly thought that $P(Y > 9) = 1 - P(Y \le 8)$. In part (c)(i), where candidates were asked for the distribution, it was not sufficient to simply quote the value of λ without also indicating that the distribution was Poisson. In part (c)(ii), the phrase 'at most 20 telephone calls' was sometimes misinterpreted, with candidates wrongly attempting to evaluate $P(T > 20)$ instead of $P(T \le 20) = 0.917$. In part (c)(iii), many candidates did not realise that P(at least 21) = 1 – P(at most 20) = 0.083. Even when this was realised, many candidates then thought that the required answer could be found by evaluating either 0.083^4 or $(0.083)^4 \times (0.917)^2$. This gave scant regard to the fact that exactly 4 one-hour periods can be achieved from the given 6 one-hour periods in

$\binom{6}{4} = 15$ different ways. It was expected that B(6, 0.083) would be used here; this was often not the case.

Q17.

This question was very well answered, with the majority of candidates gaining full marks. In part (a)(i), some candidates quoted $P(X < 3) = P(X \le 2)$ and then used the tables incorrectly, often looking up $P(X \le 3)$, presumably by mistake. Part (a)(ii) was done well by most candidates. The interpretation, in part (b), of "at least 4" was much better understood than in previous papers although there were still a few candidates who found this difficult.

Q18.

Part (a) was generally answered well. However, there were some candidates who attempted to use tables in part (a)(i), even though values are not tabulated for $\lambda = 3.5$, whilst in part (a)(iii) some candidates could not interpret 'fewer than 10' to mean '≤ 9'. It was, however, very pleasing to see fewer candidates than usual attempting to use the formula in part (a)(ii). Only the most able candidates managed to complete part (b) successfully, with the majority not realising that there were 5 ways of selecting 4 days from 5.

Part (c) was done really well by most candidates, although, in part (c)(ii), several thought that the mean had to be equal to the variance in order for a Poisson distribution to form a suitable model. Here, $\bar{x} = 9.2$ and $s^2 = 9.29$ were close enough to each other to draw this conclusion.

Q19.

In part (a)(i), the vast majority of candidates realised that the formula or their calculator had to be used in order to obtain the answer. However, there were some who tried to use Table 2 in the booklet provided by averaging the values that they found under $\lambda = 3.4$ and $\lambda = 3.6$, presumably under the misapprehension that this would give them the required value for $\lambda = 3.5$. However, most candidates did use Table 2 correctly in part (a)(ii) by realising that the most efficient way to obtain the result was by using $P(Y \ge 5) = 1 - P(Y \le 4)$. Some candidates did not seem to be able to use these tables (or perhaps even know of their existence) as they worked out $1 - P(Y = 0, 1, 2, 3 \text{ or } 4)$ by using the formula. Although they usually managed to arrive at the correct answer, this is not the most efficient way of tackling this type of question.

In part (b)(i), candidates had to indicate that the distribution was Poisson and that $\lambda = 9.5$. It was not sufficient to simply write "$\lambda = 9.5$" or just to state "Poisson". Part (b)(ii) caused most of the problems in this question. Whilst the vast majority of candidates realised correctly that $P(7 \le T \le 10) = P(T \le 10) - P(T \le 6)$, there were some who incorrectly used $P(7 \le T \le 10) = P(T \le 11) - P(T \le 6)$ or who either used the wrong value of λ or were simply careless and misread the tables. The vast majority of candidates realised that, in part (b)(iii), the correct answer could be obtained by simply evaluating (their part (b)(ii))^3. However, there were a few candidates who incorrectly thought that 3 × (their part (b)(ii)) was the way forward and did not seem at all deterred when this gave them an answer greater than one.

Q20.

Answers to part (a)(i), were invariably correct with most candidates using the statistical tables provided. A minority of candidates calculated P(X = 0) + P(X = 1) using the correct probability function for Po(0.5) and were often successful. Part (a)(ii) was done well by all but the weaker candidates who usually attempted to use tables to find P (X = 2) for λ = 2.4 and λ = 2.6 and then find the mean of the two values obtained! In part (b)(i), most candidates considered Po(3.0) and then used the formula or tables to achieve correctly the given answer of 0.224. In part (b)(ii) most candidates correctly considered $(0.224)^4$ and so scored the 2 marks available. The majority of candidates, in part(c)(ii), correctly interpreted 'T at least 18' as meaning '1 – (T ≤ 17)'.

Q21.

Part (a)(i) was answered successfully by all but the weakest candidates, with most obtaining the correct value of 0.251. Unfortunately some failed to write their answer to 3 significant figures. In part (a)(ii), too many candidates considered Po(4.5) instead of evaluating $(0.251)^3$ to obtain the correct answer of 0.0158. Most candidates realised that in order to achieve credit in part (b)(i), they had to state both the distribution (Poisson) **and** the value of its parameter (λ = 9.0). There were many excellent solutions seen to part (b)(ii) with most candidates realising that to find 'P(at least 12)' implies that P(Y ≥ 12) = 1 – P(Y ≤ 11) has to be considered. As on previous papers, candidates found it very difficult to answer the type of question posed in part (c). Many attempts did not relate to the context used in the question but simply stated general conditions under which a Poisson distribution may be appropriate.

Q22.

In part (a), the vast majority of candidates were able to state that A = 15 , with many going on to obtain full credit for the correct answer of 0.181. Candidates still had difficulty with the interpretation of 'more than' with the result that P(X>18)or P(X >18) = 1 – P(X<17) was often seen. In part (b)(i), several candidates used A = 7 instead of combining the two Poisson variables X and Y to obtain $X + Y \sim P_o$ (10).

Interpretation of 'fewer than 15' was usually stated correctly as ' < 14 ', with tables then used to obtain the required answer 0.917. In part (b)(ii), the idea of 'independence' was usually well understood although some candidates stated incorrectly that the variable was normal.

Q23.

Part (a) was usually answered well. In part (b), it was evident that most candidates realised that a description of the distribution and the value of the parameter were **both** required for full credit. The interpretation of 'at least 10' as '≥10' was usually correctly seen in part (b)(ii), with

P (X ≥ 10) = 1 – P (X ≤ 9) usually seen and Table 2 in the supplied booklet used correctly to achieve the required answer of 0.478. Part (b)(iii) was found difficult and beyond the capabilities of most candidates.

Q24.

There were many fully correct solutions to part (a) with the vast majority of these being found by the use of Table 2 in the supplied booklet. In part (b)(i), candidates realised that both a description of the distribution (Poisson) and the value of the parameter (λ = 13) were required. The first step of part (b)(ii) was answered well by most candidates, with the correct interpretation of 'at least 15' being used to give P(X ≥ 15) = 1 – P(X ≤ 14) = 0.325. Unfortunately, some candidates interpreted the phrase 'at least 15' as meaning the same as 'less than 15'. A surprising number of scripts from good candidates were seen where the solution to the second step (four successive) was missing altogether. This was the main source of error on this question.

Q25.

This question proved to be a very good source of marks with many fully correct solutions presented. In parts (a)(i) and (b)(ii), there were very many excellent solutions making good use of tables. However, some candidates used formulae to evaluate probabilities thus losing valuable time that could have been better spent on more demanding parts of the paper. Some candidates were unable to translate 'more than 8', 'fewer than 2' and 'at least eleven' into mathematically equivalent ranges.

Q26.

There were many excellent solutions to this question by candidates who realised that the total number of strikes, Y, could be represented by the distribution $P_0(2.4)$. However, it was wrongly thought by some candidates that P(3 ≤ Y ≤ 5) = P(Y ≤ 5) – P(Y ≤ 3) which inevitably led to a loss of marks. Several candidates tried to work separately with $P_0(1.1)$ $P_0(0.2)$ and consequently found this straightforward question very difficult.

Q27.

This question was well answered with many students gaining full marks.

In part (a)(i) the vast majority of candidates realised that P(Y ≥ 3) was required, with most attempting to evaluate 1 – P(Y ≤ 2). Again, candidates from some centres wasted time by evaluating individual probabilities instead of using the cumulative Poisson probability tables supplied.

In part (a)(ii) most candidates realised that they had to use the value found in part (a)(i), with most evaluating 0.6973^3 to give the correct response of 0.339. Unfortunately, some candidates attempted to use P_0 (10.8) and gained no credit.

The majority of candidates managed to gain full marks in part (b) of this question. Most used P0(7.2) to evaluate $e^{-7.2}$ whilst others correctly used P_0 (3.6) to evaluate $e^{-3.6} \times e^{-3.6}$, giving the required answer. It must be emphasised that in this question all working had to be shown since the answer was given. Quoting 0.0007 followed by 'using tables' or 'using calculator' was insufficient to gain any credit in this question.

Q28.

Nearly all candidates were able to use Poisson tables but fewer realised that the probability of 'more than one but fewer than three' is P(2). Part (b)(i) was generally well answered with fewer than expected confusing standard deviation and variance.

Q29.

This was generally well answered. Some confused standard deviation with variance in part (b)(i).

Q30.

Most candidates succeeded with part (a). In part (b), the most common incorrect answers were 7.5 or 7 .52.

Q31.

Nearly all candidates were able to identify the Poisson distribution and found this an easy starter. A few failed to use Poisson, mean 12, in part (b).

Q32.

Part (a) presented no difficulties but a surprisingly large majority calculated P(18) instead of P(18 or more) in part (b).

Q33.

This question was usually answered without difficulty.

Q34.

Part (a) was well answered. Many found part (b)(i) puzzling, giving answers such as “Normal” before using the correct distribution in part (b)(ii).

Q35.

The numerical parts were usually answered correctly but many candidates were unable to provide a convincing answer to part (c).

Q36.

This question was generally well answered. Some candidates in part (c)(i), instead of calculating the variance, assumed it was equal to the mean and so were unable to answer part (c)(ii).

Q37.

Most candidates used tables to answer this question. A few candidates made errors, such as P(2 or more) = 1 – P(2 or fewer) in part (c), but most answered it successfully. It was also well answered by those candidates who chose to calculate the probabilities but this reduced the time available to answer the rest of the paper.

Q38.

Using tables to find Poisson probabilities posed few problems. However, not all candidates used the appropriate mean when Poisson distributions were combined.